Cross-Hedging U.S. Gulf Coast Jet Fuel with New York Harbor ULSD Futures: A Minimum-Variance Hedge for an Airline Treasury Desk

Initial Term Group Project Proposal

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Topic Chosen

The world economy is heavily reliant on air transport for faster goods and passenger movement. And the cost of running this industry is heavily dependent on the price of the jet fuel. The price of jet fuel is very volatile and is affected by the price of crude oil which is itself reliant on the various global factors as we have witnessed recently in the last 8 months (the Gulf Coast jet fuel price rose from $1.82 per gallon on January 7, 2026 to $4.71 on September 15, 2026; U.S. Energy Information Administration [EIA], 2026).

For airlines, maintaining a steady supply in accordance with the real demand with stable prices is a challenge. They are exposed to the risk of price fluctuations in the market. To mitigate this risk, airlines can use futures contracts to hedge against the price volatility of jet fuel (Carter et al., 2006).

However, since there is no liquid futures contract for jet fuel, airlines often resort to cross-hedging using related contracts such as heating oil or gasoil futures (Morrell & Swan, 2006). This project aims to analyze the effectiveness of cross-hedging U.S. Gulf Coast jet fuel with NYMEX New York Harbor ULSD futures (HO), a contract that was based on heating oil until May 2013, focusing on the minimum-variance hedge ratio and its impact on reducing price risk for airlines.

The reason we chose this topic is because it is in line with what the project requires as well as it is a real world problem that is faced by airlines.

- **Real market, named decision-maker:** Gulf Coast jet fuel spot against NYMEX HO futures (42,000 gallons per contract). Hence, the decision-maker is the treasurer/risk manager of an airline.
- **Course toolkit:** Hedging the Futures with an optimal hedge ratio, doing scenario analysis, pricing, basis and roll behavior. The model will be implemented in Python.
- **Real, verifiable data collected by us:** Two public sources (mentioned in the Data Sources section), pulled and checked on November 8, 2026. We have not used a ready-made dataset with published solutions.
- **Group-specific element.** We assumed a hypothetical carrier whose fuel budget, hedge calendar and stress scenarios we will define, evaluate at a valuation date and document the data pull. The carrier buys 50 million gallons of jet fuel per quarter and hedges at horizons of one week, one month and one quarter.
- **Live demonstration.** The demo changes the estimation window, hedge horizon or hedge ratio and recomputes hedge effectiveness and margin cash needs on screen.
- **Feasible in seven weeks.** All data are already in hand, and the baseline estimate is a regression we can report by November 23.

We can all successfully learn and share with our peers and give a small snapshot of how airlines survive this highly volatile market and how they can hedge their risk using futures contracts. The project will also provide insights into the effectiveness of cross-hedging strategies and the impact of basis risk on the overall hedging performance.

Problem Statement

Let us say that an airline must buy Q gallons of jet fuel at the Gulf Coast spot price at the end of a hedge horizon τ. Its cost rises with the price, so it hedges by buying $N = hQ/42{,}000$ HO futures contracts, where h is the hedge ratio. Let ΔS and ΔF be the changes in the jet spot price and

the HO futures price over τ. The change in the airline's fuel cost per gallon, net of the futures gain, is $\Delta S - h\,\Delta F$. The minimum-variance hedge ratio is

$$h^* = \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)} = \rho\,\frac{\sigma_S}{\sigma_F},$$

(Johnson, 1960; Ederington, 1979) and hedge effectiveness is

$$\mathrm{HE} = 1 - \frac{\mathrm{Var}(\Delta S - h\,\Delta F)}{\mathrm{Var}(\Delta S)}.$$

The quantitative task is to estimate h^* on a rolling window using only data available at each decision date, and to measure the out-of-sample HE of the resulting hedge at horizons of one week, one month (21 trading days) and one quarter (63 trading days). Commodity: U.S. Gulf Coast kerosene-type jet fuel. Hedging contract: NYMEX HO front month, rolled before expiry. Sample: April 2, 1990 to November 2026. Each estimate uses a rolling 10-year window, which gives about 520 weekly, 120 monthly or 40 quarterly non-overlapping changes, and is tested on the period after the window.

Market Context & Real-World Significance

The market participants are airlines, which are the consumers, refiners and jet suppliers, which are the producers, merchants and speculators who take the other side of the hedge, and the exchange, CME Group, which clears the HO contract and sets its margin. The business decision is how many HO contracts to hold against a quarterly fuel budget, and how often to re-estimate and roll the position.

This matters the most right now primarily because airlines are a crucial industry and price of jet fuel has been extraordinarily volatile in the last 8 months. This has had a significant impact on the financial performance of both airlines, companies and the economy as a whole. Even on a short domestic route such as New York to Chicago, economy fares have risen sharply since January 2026. This is significant because this impacts businesses both by increasing the cost of supplying goods, and also by reducing the number of people who can afford the products, as well as the tourism based economy shrinking due to soaring prices of air travel, further impacting the airline industry and making the situation worse.

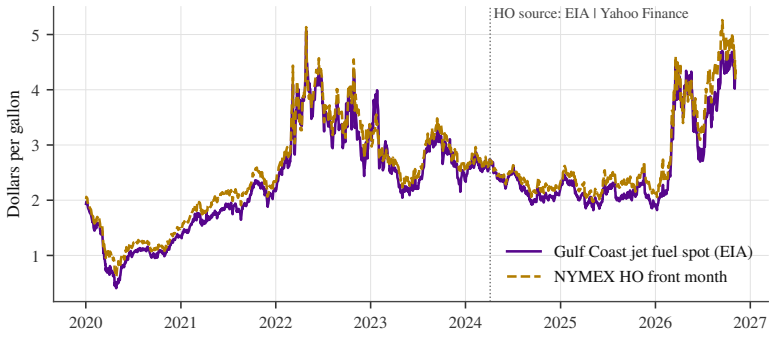
The Energy Information Administration's (EIA) Gulf Coast jet series ranged from \$1.82 per gallon on January 7, 2026 to \$4.71 on September 15, 2026, and fell 8.2% in one session, from \$4.379 on October 30 to \$4.019 on November 2 (U.S. Energy Information Administration [EIA], 2026). Moves of that size make an unhedged fuel budget fragile, and Carter et al. (2006) find that jet fuel hedging is associated with higher airline firm value, so how well a hedge actually works matters. Moves of that size also turn the futures leg into a cash problem, because daily variation margin must be paid when prices move against the position. Figure 1 shows how closely the two prices travel and how far they drift apart during shocks, which is the basis risk we measure.

Proposed Methodology

1. **Basis and roll diagnostics (futures pricing, storage and convenience yield).** We compute the basis $b_t = S_t - F_t$, study how it behaves around contract rolls, and relate its level to storage,

Figure 1

Jet Fuel Spot and NYMEX HO Front-Month Futures, 2020 to 2026



Note. Jet fuel is the U.S. Gulf Coast kerosene-type spot price (FOB). HO is the NYMEX New York Harbor front-month futures price, taken from EIA through April 5, 2024 and from Yahoo Finance afterwards. Prices are in dollars per gallon. Data through November 6, 2026 (jet fuel through November 3).

financing and convenience yield, so we can say when the cross hedge is likely to work well and when not (Geman, 2005).

2. **Minimum-variance hedge ratio (optimal hedge ratios).** We estimate h^* by an ordinary least squares regression of ΔS on ΔF, and compare it with two benchmarks: no hedge and the naive one-to-one hedge ($h = 1$).
3. **Out-of-sample evaluation.** The estimation window rolls forward and each hedge is evaluated on the period after the window. Each estimate uses only data dated on or before its decision date, so no future information enters a past decision. Every function that reads data takes an explicit decision date and uses only observations dated on or before it, and a test that adds future data to the sample checks that every past hedge ratio stays unchanged.
4. **Hedge effectiveness evaluation.** We report hedge effectiveness by horizon, estimation window and with and without roll days, with bootstrap confidence intervals, so that differences from the no-hedge and one-to-one hedges can be judged against sampling noise.
5. **Scenario and margin analysis (scenario analysis).** We replay historical stress periods (2008, 2020, 2022 and 2026) through the hedged and unhedged fuel bill and compute the variation margin the position would have required, using CME Group's published initial and maintenance margin (performance bond) levels for HO, which we will record with their retrieval date.
6. **Optional extension (price forecasting).** If the baseline is complete early, we will test whether a regression or radial basis function forecast of the basis improves the hedge.

Data Sources & Preprocessing Plan

We have a clear picture of the data we will use, and have already pulled and checked it. The two series are daily, so we will compute changes over non-overlapping horizons to avoid inflating the sample with overlapping observations. All prices are in U.S. dollars per gallon. The data sources are:

- **Jet fuel spot.** EIA, U.S. Gulf Coast kerosene-type jet fuel spot price (FOB), daily, series EER_EPJK_PF4_RGC_DPG, April 2, 1990 to November 3, 2026 (8,071 observations).

- **HO futures, EIA.** EIA, NYMEX New York Harbor No. 2 heating oil (ULSD from May 2013) front-month futures price (HO), daily, series EER_EPD2F_PE1_Y35NY_DPG, January 2, 1980 to April 5, 2024 (9,894 observations).
- **HO futures, Yahoo Finance.** Yahoo Finance, NYMEX New York Harbor ULSD front-month futures price (HO=F), daily, HO=F, April 8, 2024 to November 6, 2026 (652 observations).

Preprocessing is five steps.

- **Step 1.** We are using only the weekday close price, and we will drop any day with a missing value in either series. This leaves about 7,260 common trading days from 1990. The EIA and Yahoo Finance series overlap on 66 trading days, January 2 to April 5, 2024, which we will use to check that the two sources agree as a sanity check (they differ by at most $0.0005 per gallon).
- **Step 2.** We will compute the change in each series over the three horizons of interest: one week, one month (21 trading days) and one quarter (63 trading days). The change is the difference between the price at the end of the horizon and the price at the start of the horizon.
- **Step 3.** We will compute the basis $b_t = S_t - F_t$ for each day. A continuous front-month series jumps when the contract rolls, so we will find the roll dates from the NYMEX expiry calendar, measure the change in the futures price within a contract wherever we can, and study how the results change when the roll days are dropped or the series is back-adjusted.
- **Step 4.** The HO contract was No. 2 heating oil until May 2013 and ULSD after that, so we will repeat the estimates on the post-2013 sample alone and report any difference.
- **Step 5.** We will cross-check one estimation window in Excel.

Tools & Technical Stack

- **Python 3:** pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib, yfinance, pytest
- **Jupyter Notebook**
- **Excel**
- **Git and GitHub**

Expected Outcomes

Our goal is to provide a clear, reproducible answer to the question of how well NYMEX HO futures hedge Gulf Coast jet fuel. We will deliver a report that includes the following:

- Tables with basis and roll diagnostics, including the mean, standard deviation, and autocorrelation of the basis, as well as the frequency and magnitude of roll adjustments so as to understand the behavior of the basis and its impact on hedge effectiveness.
- Estimates of h^* with out of sample hedge effectiveness by horizon, including the one-week, one-month, and one-quarter horizons, and a comparison with the no-hedge and one-to-one benchmarks to gain a clear picture and easy to understand the effectiveness of the hedge.
- A scenario table of hedged and unhedged fuel cost with the margin each stress period would have required, including the 2008, 2020, 2022, and 2026 stress periods, to provide a clear understanding of how the hedge would have performed under known market conditions.
- A notebook that runs the whole pipeline live, allowing for easy replication and testing of different scenarios and parameters.
- A test showing that future data do not change past hedges, ensuring that the methodology is robust and reliable.

Preliminary Task Distribution Matrix

Team member	Primary role	Proposal phase	Full project
Tanishk Yadav	Hedge model lead	Writes Problem Statement and Proposed Methodology; derives the hedge-ratio and effectiveness formulas	Python hedge-ratio engine (OLS, rolling out-of-sample framework), hedge-effectiveness tables by horizon
Tianze Xia	Data and diagnostics lead	Writes Data Sources and Tools sections; EIA and Yahoo Finance pulls, access checks, preprocessing plan	Data pipeline, roll-date handling, basis and roll diagnostics, post-2013 sensitivity
Rohit Kakumani	Scenario and demo lead	Writes Market Context and Expected Outcomes; designs the stress scenarios and margin inputs	Stress scenarios and margin analysis, Excel cross-check of one window, live demo

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